

WILP, BITS Pilani

MATHEMATICAL FOUNDATIONS FOR MACHINE LEARNING · PRACTICE SET

Linear Systems by REF, RREF and Rank

Ten solved problems, with the concepts you need first

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Module 1 · Concepts, then 7 core problems, then 3 hard ones · Every step shown, nothing skipped

Prepared for the Work Integrated Learning Programmes (WILP), BITS Pilani.

How to use this set. Part I builds the ideas from scratch. Read it once, slowly. Part II has seven problems at easy and medium level. Part III has three harder ones. Every problem is stated in a **navy** box. Cover the page below it and try the problem yourself. Then read the **green** box, which works it out step by step with no gaps. The **red** boxes hold the mistake most students actually make. The **purple** box is the one line to remember.

1 Solve one right now

You already know how to do this. The matrix work is only bookkeeping.

Here are two equations:

$$x + 2y = 8 \quad (1)$$

$$3x - y = 3 \quad (2)$$

Multiply (1) by 3 and subtract it from (2). The x term dies:

$$(3x - y) - 3(x + 2y) = 3 - 3(8) \implies -7y = -21 \implies y = 3.$$

Put $y = 3$ back into (1): $x + 6 = 8$, so $x = 2$.

That is the whole method. You killed one unknown, solved for the other, then walked backwards. Everything in this document is that same move, written in a tidier way so it never gets confusing with five unknowns and four equations.

✓ Key takeaway

Gaussian elimination is one idea: use an equation to delete an unknown from the equations below it. Repeat. Then walk back up.

2 Drop the letters: the augmented matrix

The names x, y, z carry no information. Only the numbers matter, so keep only the numbers.

Take a system of m equations in n unknowns. Write it as $Ax = b$, where A is the $m \times n$ **coefficient matrix**, x is the column of unknowns, and b is the column of right-hand sides. For the warm-up above:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}, \quad x = \begin{bmatrix} x \\ y \end{bmatrix}, \quad b = \begin{bmatrix} 8 \\ 3 \end{bmatrix}.$$

Now glue b onto the right of A as one extra column. That is the **augmented matrix**, written $[A \mid b]$:

$$[A \mid b] = \left[\begin{array}{cc|c} 1 & 2 & 8 \\ 3 & -1 & 3 \end{array} \right].$$

The vertical bar is only a reminder. It says: columns left of the bar belong to unknowns, the column right of the bar is the answer side. Row i of the augmented matrix is equation i .

★ Everyday picture

Think of a recipe card. “2 cups flour, 3 eggs, 1 cup milk $\rightarrow$ 12 pancakes” carries the same content as “ $2f + 3e + 1m = 12$ ”. The words “flour”, “eggs”, “milk” are just column headings. Once everyone agrees on the column order, you can throw the headings away and keep the row of numbers.

✗ Watch out

Line up your columns before you write the matrix. If an equation is $2y + x = 5$, the row is $[1 \ 2 \mid 5]$, not $[2 \ 1 \mid 5]$. And a missing unknown is a zero, not a blank: $x + z = 4$ in three unknowns is $[1 \ 0 \ 1 \mid 4]$.

3 Three legal moves

Only three operations are allowed on rows, and each one can be undone. That is why the answer never changes.

An **elementary row operation** is one of these three:

- (R1) **Swap.** Exchange row i and row j . Written $R_i \leftrightarrow R_j$.
- (R2) **Scale.** Multiply a whole row by a nonzero number c . Written $R_i \rightarrow cR_i$.
- (R3) **Add a multiple.** Replace row i by row i plus c times row j , where $i \neq j$. Written $R_i \rightarrow R_i + cR_j$.

Each move matches a safe move on the equations themselves. Swapping is writing the equations in a different order. Scaling is multiplying both sides of one equation by the same nonzero number. Adding a multiple is adding one true equation to another true equation.

Why does the solution set survive? Because every move is **reversible**. A swap is undone by the same swap. Scaling by c is undone by scaling by $1/c$, which needs $c \neq 0$. Adding cR_j is undone by adding $-cR_j$. So the new system and the old system have exactly the same solutions. Two systems related this way are called **row equivalent**.

✗ Watch out

Two traps. First, $c = 0$ is banned in (R2): multiplying a row by zero destroys an equation and cannot be undone. Second, in (R3) you may not use row i to change itself; $R_2 \rightarrow R_2 - R_2$ is not a legal move, it just erases a row.

✓ Key takeaway

Row operations never create or destroy solutions. They only change how the same solution set is written.

4 REF: build the staircase

Push zeros into the bottom-left corner until the leading entries form a descending staircase.

In any nonzero row, the **pivot** (also called the leading entry) is the first nonzero number as you read left to right.

A matrix is in **row echelon form**, or **REF**, when all three hold:

- E1. Every all-zero row sits at the bottom.
- E2. The pivot of each row is strictly to the right of the pivot of the row above it.
- E3. Everything below a pivot, in that pivot's own column, is zero.

Condition E2 is what makes the staircase. Each step down moves you at least one column right, and sometimes more than one.

Here is a matrix in REF, with pivots shown in bold:

$$\left[\begin{array}{cccc|c} \mathbf{2} & 1 & -3 & 4 & 5 \\ 0 & 0 & \mathbf{1} & 2 & -1 \\ 0 & 0 & 0 & \mathbf{5} & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

The pivots sit in columns 1, 3, 4. Column 2 was skipped, which is allowed. The zero row is at the bottom, which is required.

A column of A that contains a pivot is a **pivot column**. A column of A with no pivot is a **free column**. This split is the single most useful thing REF gives you, and Section 7 explains why.

The intuition

REF is a triangle of zeros in the bottom-left corner. Once you have it, the last nonzero equation involves the fewest unknowns, so you can solve it first and climb upward. That climb is called **back substitution**.

Watch out

REF is not unique. Different people doing different legal moves land on different REF matrices from the same start. That is fine. What is the same for everyone is *which columns* hold the pivots, and so how many pivots there are.

5 RREF: clean above the pivots too

Make every pivot equal to 1, then clear its column above and below. Now the answer can be read straight off the page.

A matrix is in **reduced row echelon form**, or **RREF**, when it is in REF and also:

E4. Every pivot equals 1.

E5. In a pivot's column, every other entry is 0 — above it as well as below it.

Reducing the REF matrix above to RREF gives:

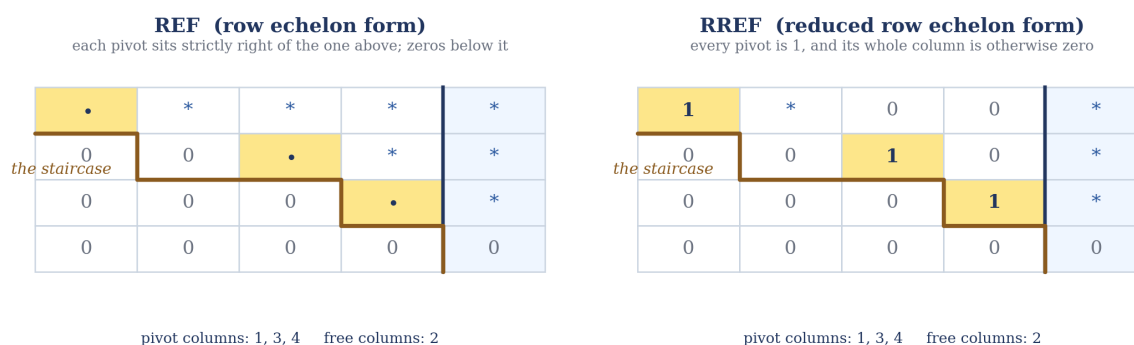
$$\left[\begin{array}{cccc|c} 1 & \frac{1}{2} & 0 & 0 & \frac{2}{5} \\ 0 & 0 & 1 & 0 & -\frac{11}{5} \\ 0 & 0 & 0 & 1 & \frac{3}{5} \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

The extra work buys you something real. In RREF each pivot variable appears in exactly one equation, with coefficient 1. So there is no back substitution left to do. You just read the answer.

Going all the way to RREF is called **Gauss–Jordan elimination**. Stopping at REF and climbing back up is called **Gaussian elimination**.

* Wait — really?

REF is not unique, but RREF is. Every matrix has exactly one RREF, no matter which legal moves you use or in which order. So two people who both reduce fully must end up with identical matrices. If your RREF differs from a friend's, one of you made an arithmetic slip.



$$\text{rank} = \text{number of pivots} = 3 \Rightarrow \text{free variables} = n - r = 4 - 3 = 1$$

Figure 1: Left: REF. The gold line is the staircase, and each pivot starts a new step further right. Right: RREF, where every pivot is 1 and its column is otherwise empty. Column 2 holds no pivot, so x_2 is free.

X Watch out

Do not chase RREF when the question only asks for rank or for consistency. Rank is visible as soon as you reach REF. Reducing further costs time and adds chances to make a

mistake. Go to RREF when you actually want the solution written out.

6 Rank: just count the pivots

The rank of a matrix is the number of pivots in its echelon form. Nothing more.

The **rank** of a matrix A , written $\text{rank}(A)$, is the number of pivots in any REF of A . Equivalently, it is the number of nonzero rows once you finish eliminating.

Three facts you will use constantly:

- $\text{rank}(A)$ counts the genuinely independent equations. Rows that are combinations of other rows collapse to zero and contribute nothing.
- For an $m \times n$ matrix, $\text{rank}(A) \leq \min(m, n)$. You cannot have more pivots than rows, and you cannot have more than one pivot per column.
- $\text{rank}([A \mid b]) \geq \text{rank}(A)$, always. Adding a column can only keep the pivot count the same or push it up by one.

That last point matters more than it looks. The extra column b can supply at most one new pivot. So there are only two possibilities: $\text{rank}([A \mid b]) = \text{rank}(A)$, or $\text{rank}([A \mid b]) = \text{rank}(A) + 1$. The whole theory of Section 7 hangs on which of these two happens.

The intuition

Rank is the amount of real information in the system. Five equations of rank 3 means two of them were repeats in disguise, and you truly have only three constraints.

Watch out

Count pivots, not nonzero entries, and count them in *echelon form*. The matrix $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ has four nonzero entries but rank 1, because $R_2 - 2R_1$ wipes the second row out.

7 The two-switch test: 0, 1, or infinitely many

Compare $\text{rank}(A)$ **with** $\text{rank}([A \mid b])$, **then compare it with** n . **Those two comparisons decide everything.**

A system is **consistent** if it has at least one solution, and **inconsistent** if it has none. Let A be $m \times n$, so n is the number of unknowns, and write $r = \text{rank}(A)$.

The rank criterion (Rouché–Capelli theorem)

- If $\text{rank}(A) < \text{rank}([A \mid b])$, the system has **no solution**.
- If $\text{rank}(A) = \text{rank}([A \mid b]) = n$, the system has **exactly one solution**.
- If $\text{rank}(A) = \text{rank}([A \mid b]) = r < n$, the system has **infinitely many solutions**, described by $n - r$ free parameters.

Read it as two switches, checked in order.

Switch 1: is it consistent? During elimination, watch for a row that reads $[0 \ 0 \ \cdots \ 0 \mid c]$ with $c \neq 0$. As an equation that says $0 = c$, which is false. One such row and the system is dead. In rank language, that row is a pivot in the b column, so $\text{rank}([A \mid b])$ exceeds $\text{rank}(A)$.

Switch 2: how many answers? Given consistency, compare r with n . If every unknown owns a pivot, each one is pinned down and the answer is unique. If some unknown has no pivot, it is free to be anything, and each choice gives a different solution. So the count of free unknowns is $n - r$, and any $n - r > 0$ means infinitely many solutions.

Two equations, two unknowns: only three things can happen

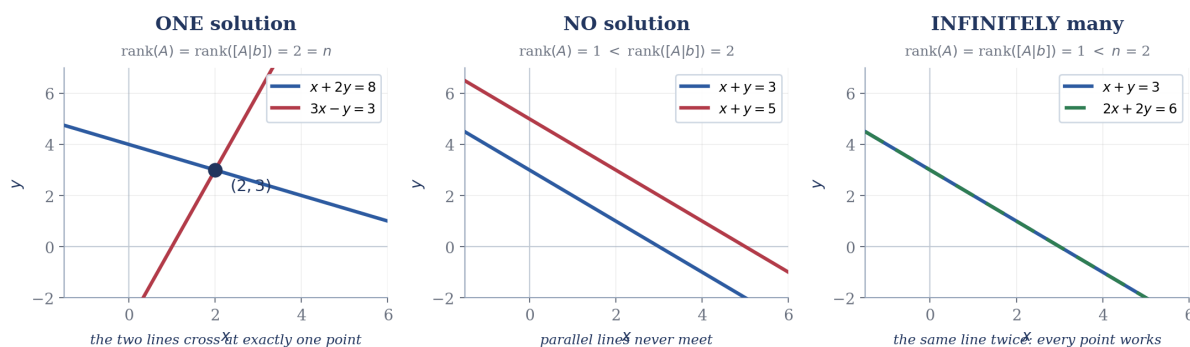


Figure 2: Two unknowns, so each equation is a line. The three cases of the rank criterion are the only three ways two lines can sit. Notice there is no picture for “exactly two solutions” — that case cannot exist.

* Wait — really?

A linear system can never have exactly 2, or 7, or 100 solutions. It has 0, or 1, or infinitely many. The reason is short: if u and v both solve $Ax = b$, then so does $u + t(v - u)$ for every real t , because $A(u + t(v - u)) = b + t(b - b) = b$. Two answers force a whole line of them.

⚠ Watch out

Never compute $\text{rank}(A)$ from one elimination and $\text{rank}([A \mid b])$ from a fresh, separate elimination. Row-reduce the augmented matrix once. Then $\text{rank}(A)$ is the pivot count in the first n columns, and $\text{rank}([A \mid b])$ is the pivot count across all $n + 1$ columns. Same picture, two readings.

8 Free variables and the shape of the answer

Every solution is one fixed answer plus anything the free variables are allowed to add.

Suppose the system is consistent with r pivots and n unknowns, and $r < n$. Here is the procedure that turns RREF into a written answer.

1. Mark the pivot columns. Their variables are **basic variables**.
2. Every remaining variable is a **free variable**. There are $n - r$ of them.
3. Give each free variable its own parameter: $x_{\text{free}} = s, t$, and so on.
4. Each row of the RREF now reads “one basic variable = number – (stuff involving the free variables)”. Write it out.
5. Collect the whole thing into vector form.

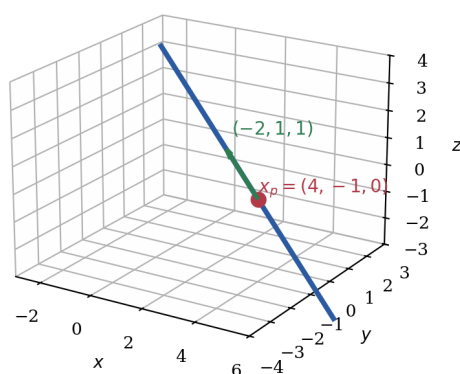
The vector form always looks like this:

$$x = \underbrace{x_p}_{\text{particular solution}} + \underbrace{s v_1 + t v_2 + \dots}_{\text{null space part}}$$

where x_p is what you get by setting every free variable to zero, and the vectors $v_1, v_2, \dots$ solve the matching homogeneous system $Ax = 0$. The set of all solutions of $Ax = 0$ is called the **null space** of A , and its dimension is $n - r$. That statement, $\text{rank}(A) + \text{nullity}(A) = n$, is the **rank–nullity theorem**.

General solution = one particular answer + everything the null space allows

$Ax = b$: one point, then slide along the line



$Ax = 0$: the same line, pushed through the origin

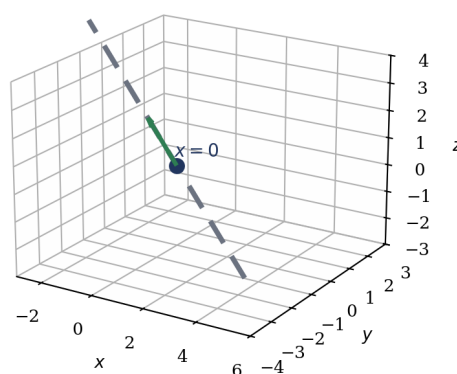


Figure 3: Left: the solutions of $Ax = b$ from Problem 4 form a line that does not pass through the origin. Right: the solutions of $Ax = 0$ form the parallel line through the origin. The two are the same direction; x_p just shifts one onto the other.

★ Everyday picture

A bus route and a bus timetable. The null space is the route: the set of directions you are allowed to move. The particular solution is where the bus starts. Every valid position is start point plus some distance along the route.

✓ Key takeaway

Number of free variables = $n - r$. That single number tells you whether the answer is a point (0 free), a line (1 free), a plane (2 free), and so on.

9 Homogeneous systems get a free pass

When $b = 0$, the system always has an answer. The only question is whether it has more than the boring one.

A system is **homogeneous** when $b = 0$, that is $Ax = 0$. Setting every unknown to zero always works, so $x = 0$ is always a solution. It is called the **trivial solution**.

This means Switch 1 can never fail. The last column of $[A \mid 0]$ is all zeros, and row operations keep it all zeros, so $\text{rank}([A \mid 0]) = \text{rank}(A)$ automatically. A homogeneous system is never inconsistent.

So only Switch 2 is live:

$$r = n \implies \text{only } x = 0; \quad r < n \implies \text{infinitely many, with } n - r \text{ free variables.}$$

One consequence is worth memorising. If a homogeneous system has more unknowns than equations, then $r \leq m < n$, so $r < n$ is forced. A homogeneous system with more unknowns than equations *always* has a nonzero solution.

✓ Key takeaway

Homogeneous: never inconsistent. Either the trivial solution alone, or a whole subspace of solutions of dimension $n - r$.

10 The recipe, and the four mistakes

One checklist covers every problem in this set.

Standard attack

1. Write $[A \mid b]$, columns lined up, missing terms as 0.
2. Row-reduce to REF. Work left to right, one pivot column at a time, clearing everything below each pivot before moving on.
3. Read $\text{rank}(A)$ (pivots in the first n columns) and $\text{rank}([A \mid b])$ (pivots overall).
4. Switch 1: if $\text{rank}(A) < \text{rank}([A \mid b])$, stop and write “no solution”.
5. Switch 2: if $\text{rank} = n$, unique. If $\text{rank} = r < n$, infinitely many with $n - r$ parameters.
6. If a solution is wanted, continue to RREF, name the free variables, and write the vector form.
7. Substitute your answer back into the *original* equations.

✗ Watch out

The four mistakes that cost the most marks:

- M1. Dividing by a parameter.** If a pivot is $k - 3$, you may not divide by it until you have said $k \neq 3$, and you must handle $k = 3$ as its own case.
- M2. Two operations that share a source row in one step.** Doing $R_2 \rightarrow R_2 - 2R_1$ and $R_1 \rightarrow R_1 + R_2$ “simultaneously” is ambiguous. One at a time, and write down what you did.
- M3. Calling $[0 \ 0 \ 0 \mid 0]$ inconsistent.** That row is harmless. Only $[0 \ 0 \ 0 \mid c]$ with $c \neq 0$ kills the system.
- M4. Counting free variables as $n - m$.** It is $n - r$. Equations that collapse to zero rows do not count.

PART II · PRACTICE SET A

The core seven: easy and medium

11 Problem 1 — A 2×2 system, done twice

Problem 1 · Easy

Solve the system below. Do it first by reducing to REF and back substituting, then again by going all the way to RREF. State $\text{rank}(A)$, $\text{rank}([A \mid b])$ and the number of unknowns, and use them to justify the number of solutions.

$$x + 2y = 8$$

$$3x - y = 3$$

What this tests. The mechanics of one elimination step, and the difference between stopping at REF and pushing on to RREF.

Worked example: Problem 1, method 1 — REF and back substitution

Step 1. Write the augmented matrix. Two unknowns, so $n = 2$.

$$[A \mid b] = \left[\begin{array}{cc|c} 1 & 2 & 8 \\ 3 & -1 & 3 \end{array} \right]$$

Step 2. Clear below the first pivot. The pivot is the 1 in position $(1, 1)$. To kill the 3 under it, use $R_2 \rightarrow R_2 - 3R_1$. Compute the new row entry by entry:

$$3 - 3(1) = 0, \quad -1 - 3(2) = -7, \quad 3 - 3(8) = -21.$$

$$\left[\begin{array}{cc|c} 1 & 2 & 8 \\ 3 & -1 & 3 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 3R_1} \left[\begin{array}{cc|c} 1 & 2 & 8 \\ 0 & -7 & -21 \end{array} \right]$$

This is REF. Pivots are 1 (column 1) and -7 (column 2).

Step 3. Read the ranks. Two pivots inside the first two columns, so $\text{rank}(A) = 2$. No pivot appears in the b column, so $\text{rank}([A \mid b]) = 2$ as well. And $n = 2$.

Step 4. Classify. Since $\text{rank}(A) = \text{rank}([A \mid b]) = 2 = n$, the system has exactly one solution.

Step 5. Back substitute. The last row says $-7y = -21$, so $y = 3$. The first row says $x + 2y = 8$, so $x = 8 - 2(3) = 2$.

Step 6. Check in the originals. $2 + 2(3) = 8 \checkmark$ and $3(2) - 3 = 3 \checkmark$.

Answer. $x = 2, y = 3$. Unique, because $\text{rank}(A) = \text{rank}([A \mid b]) = 2 = n$.

Worked example: Problem 1, method 2 — all the way to RREF

Step 1. Start from the REF above.

$$\left[\begin{array}{cc|c} 1 & 2 & 8 \\ 0 & -7 & -21 \end{array} \right]$$

Step 2. Make the second pivot equal to 1. Scale with $R_2 \rightarrow -\frac{1}{7}R_2$:

$$\left[\begin{array}{cc|c} 1 & 2 & 8 \\ 0 & 1 & 3 \end{array} \right]$$

Step 3. Clear above that pivot. Column 2 still has a 2 sitting above the pivot. Use $R_1 \rightarrow R_1 - 2R_2$:

$$1 - 0 = 1, \quad 2 - 2(1) = 0, \quad 8 - 2(3) = 2.$$

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 3 \end{array} \right]$$

This is the RREF.

Step 4. Read the answer. The rows now say $x = 2$ and $y = 3$ directly. No back substitution was needed.

Answer. Same solution, $x = 2, y = 3$. The RREF of $[A \mid b]$ is $[I \mid (2, 3)^T]$.

✓ **Key takeaway**

When the left block of the RREF turns out to be the identity matrix, the right column *is* the solution. That only happens when $\text{rank}(A) = n = m$.

12 Problem 2 — Three unknowns, one answer

Problem 2 · Easy

Use Gauss–Jordan elimination to solve

$$\begin{aligned}x + y + z &= 6 \\2x - y + z &= 3 \\x + 2y - z &= 2\end{aligned}$$

Confirm your classification with the rank criterion.

What this tests. Handling three rows without losing track, and using a row swap to dodge fractions.

Worked example: Problem 2 — full Gauss–Jordan

Step 1. Augmented matrix. Here $n = 3$.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 1 & 2 & -1 & 2 \end{array} \right]$$

Step 2. Clear column 1 below the pivot. Two operations, both using R_1 . For $R_2 \rightarrow R_2 - 2R_1$:

$$2 - 2 = 0, \quad -1 - 2 = -3, \quad 1 - 2 = -1, \quad 3 - 12 = -9.$$

For $R_3 \rightarrow R_3 - R_1$:

$$1 - 1 = 0, \quad 2 - 1 = 1, \quad -1 - 1 = -2, \quad 2 - 6 = -4.$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & -1 & -9 \\ 0 & 1 & -2 & -4 \end{array} \right]$$

Step 3. Swap to get a friendly pivot. The next pivot should sit in column 2. Taking -3 would force fractions immediately, so swap: $R_2 \leftrightarrow R_3$.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -4 \\ 0 & -3 & -1 & -9 \end{array} \right]$$

Step 4. Clear below the second pivot. Use $R_3 \rightarrow R_3 + 3R_2$:

$$-3 + 3(1) = 0, \quad -1 + 3(-2) = -7, \quad -9 + 3(-4) = -21.$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -4 \\ 0 & 0 & -7 & -21 \end{array} \right]$$

This is REF, with three pivots.

Step 5. Classify before finishing. $\text{rank}(A) = 3$, $\text{rank}([A \mid b]) = 3$, $n = 3$. All equal, so exactly one solution exists.

Step 6. Normalise the third pivot. $R_3 \rightarrow -\frac{1}{7}R_3$ gives $[0 \ 0 \ 1 \mid 3]$.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -4 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

Step 7. Clear column 3 above the pivot. Two operations: $R_2 \rightarrow R_2 + 2R_3$ gives $-4 + 2(3) = 2$ in the last slot and $-2 + 2 = 0$ in column 3. Then $R_1 \rightarrow R_1 - R_3$ gives $6 - 3 = 3$ and $1 - 1 = 0$.

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

Step 8. Clear column 2 above the pivot. $R_1 \rightarrow R_1 - R_2$ gives $3 - 2 = 1$.

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

This is the RREF.

Step 9. Check in the originals. $1 + 2 + 3 = 6 \checkmark$; $2(1) - 2 + 3 = 3 \checkmark$; $1 + 2(2) - 3 = 2 \checkmark$.

Answer. $x = 1, y = 2, z = 3$. Unique, since $\text{rank}(A) = \text{rank}([A \mid b]) = 3 = n$.

✗ Watch out

Clear column 3 *before* column 2 when going upward. If you clean column 2 first, the later column-3 operation puts new junk back into column 2 and you loop forever. Going right to left on the way up is the rule.

13 Problem 3 — When the numbers refuse

Problem 3 · Easy

Show that the following system has no solution, and identify the exact row of the reduced matrix that proves it.

$$\begin{aligned}x + y + z &= 3 \\2x + 3y + z &= 5 \\3x + 4y + 2z &= 10\end{aligned}$$

What this tests. Spotting inconsistency, and saying why in rank language rather than by feel.

Worked example: Problem 3 — detecting inconsistency

Step 1. Augmented matrix.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 2 & 3 & 1 & 5 \\ 3 & 4 & 2 & 10 \end{array} \right]$$

Step 2. Clear column 1. With $R_2 \rightarrow R_2 - 2R_1$:

$$2 - 2 = 0, \quad 3 - 2 = 1, \quad 1 - 2 = -1, \quad 5 - 6 = -1.$$

With $R_3 \rightarrow R_3 - 3R_1$:

$$3 - 3 = 0, \quad 4 - 3 = 1, \quad 2 - 3 = -1, \quad 10 - 9 = 1.$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 1 & -1 & 1 \end{array} \right]$$

Rows 2 and 3 now agree on the left of the bar but disagree on the right. That is the whole story, and the next step makes it official.

Step 3. Clear below the second pivot. $R_3 \rightarrow R_3 - R_2$:

$$0 - 0 = 0, \quad 1 - 1 = 0, \quad -1 - (-1) = 0, \quad 1 - (-1) = 2.$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 2 \end{array} \right]$$

Step 4. Read the fatal row. Row 3 is $[0 \ 0 \ 0 \mid 2]$. As an equation it says

$$0 \cdot x + 0 \cdot y + 0 \cdot z = 2, \quad \text{that is} \quad 0 = 2,$$

which no choice of x, y, z can make true.

Step 5. Say it with ranks. Inside the first three columns there are two pivots, so $\text{rank}(A) = 2$. Across all four columns there are three pivots, because row 3 has its pivot in the b column, so $\text{rank}([A \mid b]) = 3$. Since $\text{rank}(A) = 2 < 3 = \text{rank}([A \mid b])$, the system is inconsistent.

Step 6. Where it came from. Notice $R_1 + R_2 = [3 \ 4 \ 2]$ on the left, matching row 3 exactly. So the left sides satisfy $\text{eq}_1 + \text{eq}_2 = \text{eq}_3$, but the right sides give $3 + 5 = 8 \neq 10$. The equations contradict each other.

Answer. No solution. The row $[0 \ 0 \ 0 \mid 2]$ proves it, and in rank terms $\text{rank}(A) = 2 < \text{rank}([A \mid b]) = 3$.

⚠ Watch out

Once you hit a row $[0 \ \cdots \ 0 \mid c]$ with $c \neq 0$, stop. Do not carry on to RREF and do not try to write a solution. There is nothing further to compute.

14 Problem 4 — A whole line of answers

Problem 4 · Medium

Solve

$$\begin{aligned}x + y + z &= 3 \\2x + 3y + z &= 5 \\3x + 4y + 2z &= 8\end{aligned}$$

Give the general solution in parametric form and in vector form, and state how many free variables there are and why.

What this tests. The same matrix A as Problem 3 with a different b . One number changed and the answer went from “none” to “infinitely many”.

Worked example: Problem 4 — the general solution

Step 1. Augmented matrix.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 2 & 3 & 1 & 5 \\ 3 & 4 & 2 & 8 \end{array} \right]$$

Step 2. Clear column 1. $R_2 \rightarrow R_2 - 2R_1$ gives $[0 \ 1 \ -1 \ | \ -1]$ exactly as before. $R_3 \rightarrow R_3 - 3R_1$ now gives

$$3 - 3 = 0, \quad 4 - 3 = 1, \quad 2 - 3 = -1, \quad 8 - 9 = -1.$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 1 & -1 & -1 \end{array} \right]$$

Step 3. Clear below the second pivot. $R_3 \rightarrow R_3 - R_2$ wipes row 3 out completely:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

This is REF. The zero row is harmless: it says $0 = 0$.

Step 4. Classify. Two pivots in the first three columns, so $\text{rank}(A) = 2$. No pivot in the b column, so $\text{rank}([A \ | \ b]) = 2$ too. They match, so the system is consistent. But $r = 2 < 3 = n$, so there are infinitely many solutions, with

$$n - r = 3 - 2 = 1 \quad \text{free variable.}$$

Step 5. Go to RREF. Both pivots are already 1. Only column 2 needs cleaning above, using $R_1 \rightarrow R_1 - R_2$:

$$1 - 0 = 1, \quad 1 - 1 = 0, \quad 1 - (-1) = 2, \quad 3 - (-1) = 4.$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Step 6. Name the variables. Pivots sit in columns 1 and 2, so x and y are basic. Column 3 has no pivot, so z is free. Set $z = t$, where t is any real number.

Step 7. Read the rows. Row 1 says $x + 2z = 4$, so $x = 4 - 2t$. Row 2 says $y - z = -1$, so $y = -1 + t$.

Step 8. Parametric form.

$$x = 4 - 2t, \quad y = -1 + t, \quad z = t, \quad t \in \mathbb{R}.$$

Step 9. Vector form. Split the constant part from the t part:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

The first vector is a particular solution, found by setting $t = 0$. The second vector solves $Ax = 0$, and it spans the null space.

Step 10. Check in the originals, keeping t symbolic.

$$\text{eq}_1: (4 - 2t) + (-1 + t) + t = 3 + (-2t + t + t) = 3 \checkmark$$

$$\text{eq}_2: 2(4 - 2t) + 3(-1 + t) + t = 5 + (-4t + 3t + t) = 5 \checkmark$$

$$\text{eq}_3: 3(4 - 2t) + 4(-1 + t) + 2t = 8 + (-6t + 4t + 2t) = 8 \checkmark$$

Every t term cancels, which is exactly what “the direction lies in the null space” means.

Answer. Infinitely many solutions: $(x, y, z) = (4, -1, 0) + t(-2, 1, 1)$ for $t \in \mathbb{R}$. One free variable, since $n - r = 3 - 2 = 1$.

*** Wait — really?**

Problems 3 and 4 share the identical matrix A . Only b_3 moved, from 10 to 8. So $\text{rank}(A) = 2$ in both. Changing b can flip a system between “no solution” and “infinitely many”, but it can never change $\text{rank}(A)$, and it can never produce a unique solution here. With $r = 2 < n = 3$, a unique solution was impossible from the start.

15 Problem 5 — A homogeneous system

Problem 5 · Medium

Determine whether the system below has a solution other than $x = y = z = 0$. If it does, find every solution and give a basis for the null space of the coefficient matrix.

$$\begin{aligned}x + 2y + 3z &= 0 \\2x + 5y + 7z &= 0 \\3x + 7y + 10z &= 0\end{aligned}$$

What this tests. Whether you realise consistency is automatic here, so the only real question is r versus n .

Worked example: Problem 5 — non-trivial solutions and a null-space basis

Step 1. Augmented matrix, with a zero column on the right.

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 2 & 5 & 7 & 0 \\ 3 & 7 & 10 & 0 \end{array} \right]$$

That last column will stay zero through every row operation, since every operation combines zeros. So we may track only A and remember the 0.

Step 2. Clear column 1. $R_2 \rightarrow R_2 - 2R_1$:

$$2 - 2 = 0, \quad 5 - 4 = 1, \quad 7 - 6 = 1.$$

$R_3 \rightarrow R_3 - 3R_1$:

$$3 - 3 = 0, \quad 7 - 6 = 1, \quad 10 - 9 = 1.$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right]$$

Step 3. Clear below the second pivot. $R_3 \rightarrow R_3 - R_2$ gives a zero row:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Step 4. Classify. $\text{rank}(A) = 2$ and $\text{rank}([A \mid 0]) = 2$; they are equal automatically, so the system is consistent. Since $r = 2 < 3 = n$, there are $3 - 2 = 1$ free variables, so there are infinitely many solutions. So non-trivial solutions do exist.

Step 5. Reduce to RREF. Clear column 2 above the pivot with $R_1 \rightarrow R_1 - 2R_2$:

$$1 - 0 = 1, \quad 2 - 2 = 0, \quad 3 - 2 = 1.$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Step 6. Solve. Pivots in columns 1 and 2, so x and y are basic and z is free. Set $z = t$. Row 1 gives $x + z = 0$, so $x = -t$. Row 2 gives $y + z = 0$, so $y = -t$.

Step 7. Vector form.

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, \quad t \in \mathbb{R}.$$

There is no constant vector out front. A homogeneous system always has $x_p = 0$, so the answer is pure null space.

Step 8. Check. Take $t = 1$, giving $(-1, -1, 1)$:

$$-1 + 2(-1) + 3(1) = 0 \checkmark, \quad -2 - 5 + 7 = 0 \checkmark, \quad -3 - 7 + 10 = 0 \checkmark$$

Step 9. Rank-nullity check. $\text{rank}(A) = 2$, $\text{nullity}(A) = 1$, and $2 + 1 = 3 = n \checkmark$.

Answer. Non-trivial solutions exist. Every solution is $t(-1, -1, 1)$, and a basis for the null space is $\{(-1, -1, 1)\}$.

⚠ Watch out

Writing the null space as “ $(-1, -1, 1)$ ” alone is incomplete. The null space is the whole *set* $\{t(-1, -1, 1) : t \in \mathbb{R}\}$. The single vector is a *basis* for it. Also note the trivial solution is still in there: it is the case $t = 0$.

16 Problem 6 — Rank of a matrix on its own

Problem 6 · Medium

For the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 1 & 3 & 5 & 7 \\ 2 & 5 & 8 & 11 \end{bmatrix}$$

find $\text{rank}(A)$. Then decide whether $Ax = 0$ has a non-trivial solution, and if so, find a basis for the null space. Verify the rank–nullity theorem.

What this tests. Rank as a property of a matrix, without any b attached, plus the link from rank straight to the null space.

Worked example: Problem 6 — rank, null space, rank–nullity

Step 1. Look before you compute. Row 2 is exactly $2 \times$ row 1. Also row 1 plus row 3 is $[2 \ 5 \ 8 \ 11]$, which is row 4. So two of the four rows are combinations of the others, and the rank should come out as 2. The elimination below confirms it.

Step 2. Clear column 1. Three operations, all using R_1 :

$$R_2 \rightarrow R_2 - 2R_1 : [2 - 2, 4 - 4, 6 - 6, 8 - 8] = [0 \ 0 \ 0 \ 0]$$

$$R_3 \rightarrow R_3 - R_1 : [1 - 1, 3 - 2, 5 - 3, 7 - 4] = [0 \ 1 \ 2 \ 3]$$

$$R_4 \rightarrow R_4 - 2R_1 : [2 - 2, 5 - 4, 8 - 6, 11 - 8] = [0 \ 1 \ 2 \ 3]$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \end{bmatrix}$$

Step 3. Move the zero row down. REF requires zero rows at the bottom, so swap $R_2 \leftrightarrow R_3$:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 \end{bmatrix}$$

Step 4. Clear the last row. $R_4 \rightarrow R_4 - R_2$ gives another zero row:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

This is REF. Two nonzero rows, so two pivots, in columns 1 and 2.

Step 5. State the rank. $\text{rank}(A) = 2$.

Step 6. Decide on non-trivial solutions. Here $n = 4$ and $r = 2$, so $n - r = 2 > 0$. A homogeneous system with a rank below the number of unknowns always has non-trivial solutions. So yes.

Step 7. Reduce to RREF. Both pivots are already 1. Clear column 2 above with $R_1 \rightarrow R_1 - 2R_2$:

$$1 - 0 = 1, \quad 2 - 2 = 0, \quad 3 - 4 = -1, \quad 4 - 6 = -2.$$

$$\begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Step 8. Write the equations. Pivot columns 1 and 2, so x_1 and x_2 are basic; columns 3 and 4 have no pivots, so x_3 and x_4 are free.

$$x_1 - x_3 - 2x_4 = 0, \quad x_2 + 2x_3 + 3x_4 = 0.$$

Step 9. Introduce parameters. Put $x_3 = s$ and $x_4 = t$. Then

$$x_1 = s + 2t, \quad x_2 = -2s - 3t.$$

Step 10. Split into two vectors. Group the s terms and the t terms:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = s \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \end{bmatrix}$$

Each vector comes from setting one free variable to 1 and the other to 0. That is why the pattern of 1 and 0 appears in the bottom two slots, and it also proves the two vectors are independent.

Step 11. Check both against a row of the original. Use row 4, $[2 \ 5 \ 8 \ 11]$:

$$2(1) + 5(-2) + 8(1) + 11(0) = 2 - 10 + 8 = 0 \checkmark$$

$$2(2) + 5(-3) + 8(0) + 11(1) = 4 - 15 + 11 = 0 \checkmark$$

Step 12. Rank-nullity. $\text{nullity}(A) = 2$ because the basis has two vectors, and

$$\text{rank}(A) + \text{nullity}(A) = 2 + 2 = 4 = n \checkmark$$

Answer. $\text{rank}(A) = 2$. Non-trivial solutions exist, and a null-space basis is $\{(1, -2, 1, 0), (2, -3, 0, 1)\}$, so $\text{nullity}(A) = 2$ and $2 + 2 = 4$.

✓ Key takeaway

Free variables and null-space basis vectors are the same thing counted twice. Set one free variable to 1 and the rest to 0, and you have read off one basis vector. Repeat for each free variable.

17 Problem 7 — More unknowns than equations

Problem 7 · Medium

Solve the system

$$x_1 + 2x_2 - x_3 + x_4 = 1$$

$$2x_1 + 4x_2 + x_3 - x_4 = 5$$

Identify the pivot and free variables, and write the complete solution in vector form.

What this tests. That the free variables are decided by the pivot positions, not by which letters look convenient.

Worked example: Problem 7 — two equations, four unknowns

Step 1. Augmented matrix. Here $m = 2$ and $n = 4$.

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 1 \\ 2 & 4 & 1 & -1 & 5 \end{array} \right]$$

Step 2. Predict the outcome first. Rank cannot exceed the number of rows, so $r \leq 2 < 4 = n$. A unique solution is impossible. The only two options are “no solution” or “infinitely many”.

Step 3. Clear column 1. $R_2 \rightarrow R_2 - 2R_1$:

$$2 - 2 = 0, \quad 4 - 4 = 0, \quad 1 - (-2) = 3, \quad -1 - 2 = -3, \quad 5 - 2 = 3.$$

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 1 \\ 0 & 0 & 3 & -3 & 3 \end{array} \right]$$

Column 2 came out entirely zero below the first row. So the second pivot is not in column 2, it is in column 3.

Step 4. Scale the second row. $R_2 \rightarrow \frac{1}{3}R_2$:

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 1 \\ 0 & 0 & 1 & -1 & 1 \end{array} \right]$$

This is REF, with pivots in columns 1 and 3.

Step 5. Classify. $\text{rank}(A) = 2 = \text{rank}([A \mid b])$, so the system is consistent. And $n - r = 4 - 2 = 2$, so there are two free variables and infinitely many solutions.

Step 6. Reduce to RREF. Clear column 3 above the second pivot with $R_1 \rightarrow R_1 + R_2$:

$$1 + 0 = 1, \quad 2 + 0 = 2, \quad -1 + 1 = 0, \quad 1 + (-1) = 0, \quad 1 + 1 = 2.$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 0 & 2 \\ 0 & 0 & 1 & -1 & 1 \end{array} \right]$$

Step 7. Name the variables. Pivots are in columns 1 and 3, so x_1 and x_3 are basic. Columns 2 and 4 have no pivot, so x_2 and x_4 are free. Put $x_2 = s$ and $x_4 = t$.

Step 8. Read the rows. Row 1 says $x_1 + 2x_2 = 2$, so $x_1 = 2 - 2s$. Row 2 says $x_3 - x_4 = 1$, so $x_3 = 1 + t$.

Step 9. Vector form. Collect constants, then the s part, then the t part:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

Step 10. Check with symbols kept in.

$$\text{eq}_1 : (2 - 2s) + 2s - (1 + t) + t = 2 - 1 = 1 \checkmark$$

$$\text{eq}_2 : 2(2 - 2s) + 4s + (1 + t) - t = 4 - 4s + 4s + 1 = 5 \checkmark$$

Answer. Infinitely many solutions with two free variables: $x = (2, 0, 1, 0) + s(-2, 1, 0, 0) + t(0, 0, 1, 1)$.

⚠ Watch out

The tempting mistake is to declare x_3 and x_4 free “because they are the last two”. Look at the RREF. The pivots landed in columns 1 and 3, so the free ones are x_2 and x_4 . Free variables are the columns without pivots, full stop.

PART III · PRACTICE SET B

The tough three

18 Problem 8 — A system with two dials

Problem 8 · Hard

Consider the system

$$\begin{aligned}x + y + z &= 6 \\x + 2y + 3z &= 14 \\x + 2y + az &= b\end{aligned}$$

where a and b are real constants. Find all values of a and b for which the system has (i) a unique solution, (ii) no solution, (iii) infinitely many solutions. In case (i) give the solution in terms of a and b . In case (iii) give the general solution.

What this tests. Eliminating while a pivot might secretly be zero. This is the classic exam problem, and the classic way to lose marks is to divide by $a - 3$ without checking whether it is zero.

Worked example: Problem 8 — the three cases

Step 1. Augmented matrix. Here $n = 3$.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 14 \\ 1 & 2 & a & b \end{array} \right]$$

Step 2. Clear column 1. The pivot is a plain 1, and it does not depend on a or b , so this step is always safe. $R_2 \rightarrow R_2 - R_1$:

$$1 - 1 = 0, \quad 2 - 1 = 1, \quad 3 - 1 = 2, \quad 14 - 6 = 8.$$

$R_3 \rightarrow R_3 - R_1$:

$$1 - 1 = 0, \quad 2 - 1 = 1, \quad a - 1, \quad b - 6.$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 1 & a - 1 & b - 6 \end{array} \right]$$

Step 3. Clear below the second pivot. The second pivot is 1, again free of a and b , so $R_3 \rightarrow R_3 - R_2$ is safe:

$$1 - 1 = 0, \quad (a - 1) - 2 = a - 3, \quad (b - 6) - 8 = b - 14.$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & a - 3 & b - 14 \end{array} \right]$$

Step 4. Stop and look at row 3. Everything now depends on the single row

$$[0 \ 0 \ a - 3 \mid b - 14],$$

which as an equation reads $(a - 3)z = b - 14$. Whether this row carries a pivot, carries a contradiction, or vanishes decides the whole problem. Do not divide by $a - 3$ yet.

Step 5. Case (i): $a \neq 3$. Then $a - 3 \neq 0$, so row 3 has a pivot in column 3. There are three pivots in the first three columns, giving $\text{rank}(A) = 3$, and none in the b column, so $\text{rank}([A \mid b]) = 3$ as well. With $3 = 3 = n$, the solution is unique, whatever b is. Now division is legal:

$$z = \frac{b - 14}{a - 3}.$$

Back substitute into row 2, $y + 2z = 8$:

$$y = 8 - 2z = 8 - \frac{2(b - 14)}{a - 3}.$$

Back substitute into row 1, $x + y + z = 6$:

$$\begin{aligned} x &= 6 - y - z \\ &= 6 - \left(8 - \frac{2(b - 14)}{a - 3}\right) - \frac{b - 14}{a - 3} \\ &= -2 + \frac{2(b - 14)}{a - 3} - \frac{b - 14}{a - 3} \\ &= -2 + \frac{b - 14}{a - 3}. \end{aligned}$$

Step 6. Spot-check case (i) with numbers. Take $a = 4$, $b = 15$. Then $z = 1/1 = 1$, $y = 8 - 2 = 6$, $x = -2 + 1 = -1$. Test all three originals:

$$-1 + 6 + 1 = 6 \checkmark, \quad -1 + 12 + 3 = 14 \checkmark, \quad -1 + 12 + 4 = 15 \checkmark$$

Step 7. Case (ii): $a = 3$ and $b \neq 14$. Row 3 becomes $[0 \ 0 \ 0 \mid b - 14]$ with $b - 14 \neq 0$. That is the equation $0 = b - 14$, which is false. Column 3 has lost its pivot, so $\text{rank}(A) = 2$, while the b column has gained one, so $\text{rank}([A \mid b]) = 3$. Since $2 < 3$, there is no solution.

Step 8. Case (iii): $a = 3$ and $b = 14$. Row 3 becomes $[0 \ 0 \ 0 \mid 0]$, which is harmless. Now $\text{rank}(A) = \text{rank}([A \mid b]) = 2$, and $2 < 3 = n$, so there are infinitely many solutions with $3 - 2 = 1$ free variable.

Step 9. Solve case (iii). The matrix is

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Pivots in columns 1 and 2, so z is free. Put $z = t$. Row 1 gives $x - z = -2$, so $x = -2 + t$. Row 2 gives $y + 2z = 8$, so $y = 8 - 2t$.

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2 \\ 8 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

Step 10. Check case (iii) symbolically. With $a = 3$, $b = 14$ the third equation is identical to the second, so only two need checking:

$$\begin{aligned} (-2 + t) + (8 - 2t) + t &= 6 + (t - 2t + t) = 6 \checkmark \\ (-2 + t) + 2(8 - 2t) + 3t &= 14 + (t - 4t + 3t) = 14 \checkmark \end{aligned}$$

Answer. Unique when $a \neq 3$ (any b), with $z = \frac{b-14}{a-3}$, $y = 8 - \frac{2(b-14)}{a-3}$, $x = -2 + \frac{b-14}{a-3}$. No solution when $a = 3, b \neq 14$. Infinitely many when $a = 3, b = 14$, namely $(-2, 8, 0) + t(1, -2, 1)$.

✗ Watch out

Note which parameter controls which switch. Here a alone decides $\text{rank}(A)$, because a lives on the left of the bar. Then b decides, only in the knife-edge case $a = 3$, whether the system is consistent. That division of labour is typical: parameters inside A control the rank, and parameters inside b control consistency.

*** Wait — really?**

At $a = 3$ the answer does not change gradually, it changes character. For $a = 3.001$ you get one point. At $a = 3$ exactly you get either nothing at all or an entire line, depending on b . And look at the unique solution as $a \rightarrow 3$ with $b \neq 14$: the fraction $\frac{b-14}{a-3}$ blows up. The point runs off to infinity, and in the limit there is nowhere for it to be.

19 Problem 9 — Four equations, five unknowns

Problem 9 · Hard

Solve

$$\begin{aligned}x_1 + 2x_2 - x_3 + 3x_4 + x_5 &= 3 \\2x_1 + 4x_2 - x_3 + 4x_4 + 3x_5 &= 5 \\3x_1 + 6x_2 - 2x_3 + 7x_4 + 4x_5 &= 8 \\x_1 + 2x_2 + x_3 &\quad + 5x_5 = 2\end{aligned}$$

Give $\text{rank}(A)$, $\text{rank}([A \mid b])$, the number of free variables, the general solution in vector form, and a basis for the null space of A .

What this tests. A full-size elimination where one equation is redundant, the pivots skip columns, and the answer is a two-parameter family. Note the missing x_4 term in the last equation: it enters the matrix as a zero.

Worked example: Problem 9, part 1 — reduce to REF

Step 1. Augmented matrix. Here $m = 4$ and $n = 5$.

$$\left[\begin{array}{ccccc|c} 1 & 2 & -1 & 3 & 1 & 3 \\ 2 & 4 & -1 & 4 & 3 & 5 \\ 3 & 6 & -2 & 7 & 4 & 8 \\ 1 & 2 & 1 & 0 & 5 & 2 \end{array} \right]$$

Step 2. Clear column 1. Three operations, all against R_1 :

$$R_2 \rightarrow R_2 - 2R_1 : [0, 0, -1 + 2, 4 - 6, 3 - 2 \mid 5 - 6] = [0, 0, 1, -2, 1 \mid -1]$$

$$R_3 \rightarrow R_3 - 3R_1 : [0, 0, -2 + 3, 7 - 9, 4 - 3 \mid 8 - 9] = [0, 0, 1, -2, 1 \mid -1]$$

$$R_4 \rightarrow R_4 - R_1 : [0, 0, 1 + 1, 0 - 3, 5 - 1 \mid 2 - 3] = [0, 0, 2, -3, 4 \mid -1]$$

$$\left[\begin{array}{ccccc|c} 1 & 2 & -1 & 3 & 1 & 3 \\ 0 & 0 & 1 & -2 & 1 & -1 \\ 0 & 0 & 1 & -2 & 1 & -1 \\ 0 & 0 & 2 & -3 & 4 & -1 \end{array} \right]$$

Column 2 is now entirely zero below row 1. So no pivot can sit in column 2, and the next pivot is in column 3.

Step 3. Clear below the second pivot, in column 3. Two operations against R_2 :

$$R_3 \rightarrow R_3 - R_2 : [0, 0, 0, 0, 0 \mid 0]$$

$$R_4 \rightarrow R_4 - 2R_2 : [0, 0, 0, -3 + 4, 4 - 2 \mid -1 + 2] = [0, 0, 0, 1, 2 \mid 1]$$

$$\left[\begin{array}{ccccc|c} 1 & 2 & -1 & 3 & 1 & 3 \\ 0 & 0 & 1 & -2 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 1 \end{array} \right]$$

Step 4. Move the zero row to the bottom. REF demands it, so $R_3 \leftrightarrow R_4$:

$$\left[\begin{array}{ccccc|c} 1 & 2 & -1 & 3 & 1 & 3 \\ 0 & 0 & 1 & -2 & 1 & -1 \\ 0 & 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

This is REF. Pivots sit in columns 1, 3, 4.

Step 5. Read the ranks and classify. Three pivots inside the first five columns, so $\text{rank}(A) = 3$. No pivot in the b column, so $\text{rank}([A \mid b]) = 3$ too. They match, so the system is consistent. And

$$n - r = 5 - 3 = 2,$$

so there are two free variables and infinitely many solutions.

Step 6. Where the redundancy was. Row 3 of the original equals row 1 plus row 2, on both sides: $3 + 5 = 8$ matches. So the third equation carried no new information, which is why a zero row appeared.

 Worked example: Problem 9, part 2 — RREF and the general solution

Step 1. Clear column 4, the rightmost pivot column, first. The pivot is the 1 in row 3. Rows 1 and 2 have entries 3 and -2 above it.

$$R_2 \rightarrow R_2 + 2R_3 : [0, 0, 1, -2 + 2, 1 + 4 \mid -1 + 2] = [0, 0, 1, 0, 5 \mid 1]$$

$$R_1 \rightarrow R_1 - 3R_3 : [1, 2, -1, 3 - 3, 1 - 6 \mid 3 - 3] = [1, 2, -1, 0, -5 \mid 0]$$

$$\left[\begin{array}{ccccc|c} 1 & 2 & -1 & 0 & -5 & 0 \\ 0 & 0 & 1 & 0 & 5 & 1 \\ 0 & 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Step 2. Now clear column 3. Row 1 has -1 above that pivot, so $R_1 \rightarrow R_1 + R_2$:

$$[1, 2, -1 + 1, 0, -5 + 5 \mid 0 + 1] = [1, 2, 0, 0, 0 \mid 1]$$

$$\left[\begin{array}{ccccc|c} 1 & 2 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 5 & 1 \\ 0 & 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

This is the RREF. Column 1 needs nothing, since its pivot already sits alone.

Step 3. Name the variables. Pivot columns are 1, 3, 4, so x_1, x_3, x_4 are basic. Columns 2 and 5 have no pivot, so x_2 and x_5 are free. Put $x_2 = s$ and $x_5 = t$.

Step 4. Read one row at a time.

$$\text{Row 1: } x_1 + 2x_2 = 1 \quad \implies \quad x_1 = 1 - 2s$$

$$\text{Row 2: } x_3 + 5x_5 = 1 \quad \implies \quad x_3 = 1 - 5t$$

$$\text{Row 3: } x_4 + 2x_5 = 1 \quad \implies \quad x_4 = 1 - 2t$$

Step 5. Assemble the vector form.

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 5 \\ -5 \\ -2 \\ 1 \end{bmatrix}$$

Step 6. Identify the three pieces. Setting $s = t = 0$ gives the particular solution $x_p = (1, 0, 1, 1, 0)$. The two vectors multiplying s and t solve $Ax = 0$, so they form a basis of the null space.

Step 7. Check the particular solution in all four originals.

$$\begin{array}{ll} 1 + 0 - 1 + 3 + 0 = 3 \checkmark & 2 + 0 - 1 + 4 + 0 = 5 \checkmark \\ 3 + 0 - 2 + 7 + 0 = 8 \checkmark & 1 + 0 + 1 + 0 + 0 = 2 \checkmark \end{array}$$

Step 8. Check a general point. Take $s = t = 1$, giving $x = (-1, 1, -4, -1, 1)$:

$$\begin{array}{l} \text{eq}_1 : -1 + 2 + 4 - 3 + 1 = 3 \checkmark \\ \text{eq}_2 : -2 + 4 + 4 - 4 + 3 = 5 \checkmark \\ \text{eq}_3 : -3 + 6 + 8 - 7 + 4 = 8 \checkmark \\ \text{eq}_4 : -1 + 2 - 4 + 0 + 5 = 2 \checkmark \end{array}$$

Step 9. Rank-nullity. $\text{rank}(A) + \text{nullity}(A) = 3 + 2 = 5 = n \checkmark$.

Answer. $\text{rank}(A) = \text{rank}([A \mid b]) = 3$, two free variables, and $x = (1, 0, 1, 1, 0) + s(-2, 1, 0, 0, 0) + t(0, 0, -5, -2, 1)$. Null-space basis: $\{(-2, 1, 0, 0, 0), (0, 0, -5, -2, 1)\}$.

⚠ Watch out

Four equations did not give four pivots. Do not assume $\text{rank}(A) = m$. Here one equation was the sum of two others, so the rank was 3, not 4. Equally, do not assume the free variables are the last ones: they are x_2 and x_5 .

20 Problem 10 — Which right-hand sides are allowed?

Problem 10 · Hard

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 1 & 3 & 5 \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

Find $\text{rank}(A)$. Then find the exact condition on b_1, b_2, b_3 under which $Ax = b$ is consistent. When the condition holds, give the general solution in terms of b_1, b_2, b_3 . Finally, verify your condition on $b = (1, 3, 2)$.

What this tests. Running the elimination with unknown symbols on the right-hand side. This is the reverse of Problem 8: there the left side carried the parameters, here the right side does.

Worked example: Problem 10 — the consistency condition

Step 1. Augmented matrix, with symbols on the right.

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & b_1 \\ 2 & 5 & 8 & b_2 \\ 1 & 3 & 5 & b_3 \end{array} \right]$$

Treat b_1, b_2, b_3 as ordinary numbers whose values you do not know. Every row operation acts on them just like on any other entry.

Step 2. Clear column 1. $R_2 \rightarrow R_2 - 2R_1$:

$$2 - 2 = 0, \quad 5 - 4 = 1, \quad 8 - 6 = 2, \quad b_2 - 2b_1.$$

$R_3 \rightarrow R_3 - R_1$:

$$1 - 1 = 0, \quad 3 - 2 = 1, \quad 5 - 3 = 2, \quad b_3 - b_1.$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & b_1 \\ 0 & 1 & 2 & b_2 - 2b_1 \\ 0 & 1 & 2 & b_3 - b_1 \end{array} \right]$$

Step 3. Clear below the second pivot. $R_3 \rightarrow R_3 - R_2$. The left side vanishes completely, and the right side becomes

$$(b_3 - b_1) - (b_2 - 2b_1) = b_3 - b_1 - b_2 + 2b_1 = b_1 - b_2 + b_3.$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & b_1 \\ 0 & 1 & 2 & b_2 - 2b_1 \\ 0 & 0 & 0 & b_1 - b_2 + b_3 \end{array} \right]$$

Step 4. Read the rank of A . Only two pivots survive on the left, in columns 1 and 2, so $\text{rank}(A) = 2$, whatever b is. Column 3 has no pivot, so z will be free.

Step 5. Impose consistency. Row 3 reads $0 = b_1 - b_2 + b_3$. If the right side is nonzero, that row carries a pivot in the b column, making $\text{rank}([A \mid b]) = 3 > 2 = \text{rank}(A)$, and the system is dead. So the system is consistent exactly when

$$b_1 - b_2 + b_3 = 0$$

Step 6. Note there is no third case. When the condition holds, $\text{rank}(A) = \text{rank}([A \mid b]) = 2 < 3 = n$, so the answer is infinitely many solutions with one free variable. A unique solution can never occur, for any b , because $\text{rank}(A) = 2$ is fixed at 2 and can never reach $n = 3$.

Step 7. Reduce to RREF, assuming the condition holds. $R_1 \rightarrow R_1 - 2R_2$:

$$1, \quad 2 - 2 = 0, \quad 3 - 4 = -1, \quad b_1 - 2(b_2 - 2b_1) = 5b_1 - 2b_2.$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 5b_1 - 2b_2 \\ 0 & 1 & 2 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Step 8. Write the general solution. Set $z = t$. Row 1 gives $x - z = 5b_1 - 2b_2$, and row 2 gives $y + 2z = b_2 - 2b_1$:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5b_1 - 2b_2 \\ b_2 - 2b_1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

The direction vector $(1, -2, 1)$ carries no b at all. That is expected: the null space belongs to A alone, so changing b slides the solution line around, but never turns it.

Step 9. Verify with $b = (1, 3, 2)$. Check the condition first: $1 - 3 + 2 = 0 \checkmark$, so it should be consistent. The formula gives

$$x = 5(1) - 2(3) + t = -1 + t, \quad y = 3 - 2(1) - 2t = 1 - 2t, \quad z = t.$$

Test at $t = 0$, so $(x, y, z) = (-1, 1, 0)$:

$$-1 + 2 + 0 = 1 \checkmark, \quad -2 + 5 + 0 = 3 \checkmark, \quad -1 + 3 + 0 = 2 \checkmark$$

Test at $t = 1$, so $(x, y, z) = (0, -1, 1)$:

$$0 - 2 + 3 = 1 \checkmark, \quad 0 - 5 + 8 = 3 \checkmark, \quad 0 - 3 + 5 = 2 \checkmark$$

Step 10. Verify a failing case too. Take $b = (1, 3, 5)$, where $1 - 3 + 5 = 3 \neq 0$. Row 3 becomes $[0 \ 0 \ 0 \mid 3]$, so the system has no solution, as the condition predicted.

Answer. $\text{rank}(A) = 2$. Consistent exactly when $b_1 - b_2 + b_3 = 0$, and then $(x, y, z) = (5b_1 - 2b_2, b_2 - 2b_1, 0) + t(1, -2, 1)$. Never unique, for any b .

The intuition

The condition $b_1 - b_2 + b_3 = 0$ has a clean picture behind it. Since $\text{rank}(A) = 2$, the three columns of A span only a plane inside $\mathbb{R}^3$, not the whole space. And Ax is a combination of those columns, so Ax can only ever land on that plane. The system is solvable exactly when b already lies there. The vector $(1, -1, 1)$ is the normal to the plane, which is why the condition reads $b_1 - b_2 + b_3 = 0$. You can see it directly: each column of A dotted with $(1, -1, 1)$ gives $1 - 2 + 1 = 0$, $2 - 5 + 3 = 0$ and $3 - 8 + 5 = 0$.

✓ Key takeaway

A parameter on the left of the bar controls $\text{rank}(A)$ and so the number of solutions. A parameter on the right of the bar controls only whether any solution exists.

PART IV

Self-test and pocket card

21 Self-test: eight quick checks

Answer each one in a sentence, then turn the page upside down.

- Q1.** A system has 6 unknowns, $\text{rank}(A) = 4$ and $\text{rank}([A \mid b]) = 4$. How many solutions, and how many free variables?
 Consistent, since the ranks match. Free variables = $n - r = 6 - 4 = 2$, so infinitely many solutions.
- Q2.** Can a linear system have exactly three solutions?
 No. If u and v both work, so does $u + t(v - u)$ for every real t . Two answers force infinitely many. Only 0, 1 or infinity are possible.
- Q3.** A is 5×3 with $\text{rank}(A) = 3$. What are the possible solution counts for $Ax = b$?
 Zero or one only. If consistent then $r = 3 = n$, so unique. If $\text{rank}([A \mid b]) = 4$, none. Infinitely many is impossible.
- Q4.** A homogeneous system has 3 equations and 5 unknowns. Must it have a nonzero solution?
 Yes. $r \leq m = 3 < 5 = n$, so $n - r \geq 2$ free variables exist. More unknowns than equations always forces non-trivial solutions.
- Q5.** The RREF of $[A \mid b]$ contains the row $[0 \ 0 \ 0 \mid 0]$. Is the system inconsistent?
 No. That row just says $0 = 0$ and can be ignored. Only a row $[0 \ \dots \ 0 \mid c]$ with $c \neq 0$ kills the system.
- Q6.** Two students reduce the same matrix and get different REF matrices. Is one of them wrong?
 Not necessarily. REF is not unique. But the pivot positions, and so the rank, must agree. Their RREF matrices must be identical.
- Q7.** If $\text{rank}(A) = \text{rank}([A \mid b])$, what does that tell you about b ?
 That b is a combination of the columns of A , that is, b lies in the column space of A . Adding it as a column brings no new pivot.
- Q8.** A is 4×4 with $\text{rank}(A) = 4$. What can you say about $Ax = b$ and about $Ax = 0$?
 $Ax = b$ has exactly one solution for every b , and $Ax = 0$ has only $x = 0$. Here $r = n = 4$, so no free variables and A is invertible.

22 Pocket card

Everything in this document, on one page.

The classification table

What you see	Meaning	Answer
$\text{rank}(A) < \text{rank}([A \mid b])$	a row $[0 \cdots 0 \mid c], c \neq 0$	no solution
$\text{rank}(A) = \text{rank}([A \mid b]) = n$	every unknown has a pivot	exactly one
$\text{rank}(A) = \text{rank}([A \mid b]) = r < n$	$n - r$ columns lack pivots	infinitely many

The facts

- Pivot = first nonzero entry in a row. Rank = number of pivots.
- REF: staircase, zeros below each pivot, zero rows last. Not unique.
- RREF: also pivots = 1 with zeros above them. Unique.
- Free variables = $n - r$. Pivot columns give basic variables.
- General solution = particular solution + null space.
- Rank-nullity: $\text{rank}(A) + \text{nullity}(A) = n$.
- Homogeneous $Ax = 0$ is never inconsistent. Only $x = 0$ if $r = n$, otherwise a subspace of dimension $n - r$.
- More unknowns than equations, and homogeneous: non-trivial solutions guaranteed.
- $\text{rank}(A) \leq \min(m, n)$, and $\text{rank}([A \mid b])$ is either $\text{rank}(A)$ or $\text{rank}(A) + 1$.
- Square A : $\text{rank}(A) = n$ is the same as $\det(A) \neq 0$, the same as A invertible.

The order of work

Augment $\rightarrow$ REF $\rightarrow$ read both ranks $\rightarrow$ switch 1 (consistent?) $\rightarrow$ switch 2 (unique or many?) $\rightarrow$ RREF if a solution is wanted $\rightarrow$ substitute back into the originals.

✓ Key takeaway

Two comparisons settle every linear system you will ever meet. Compare $\text{rank}(A)$ with $\text{rank}([A \mid b])$ to learn whether an answer exists. Compare $\text{rank}(A)$ with n to learn how many.